MORPHISMS INTO X GIVE POINTS OF THE HILBERT FUNCTOR VIA GRAPHS

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Allowed standard prerequisites (editorial).

- Lemma spaces-morphisms-lemma-base-change-separated (canonical tag 03KL). cited at proof line 3377. Separatedness stable.
- Lemma spaces-morphisms-lemma-base-change-flat (canonical tag 03MO). cited at proof line 3378. Flatness stable.
- Lemma spaces-morphisms-lemma-base-change-proper (canonical tag 04WP). cited at proof line 3379. Properness stable.
- Lemma spaces-morphisms-lemma-base-change-finite-presentation (canonical tag 03XR). cited at proof line 3380. fp stable.
- Lemma spaces-morphisms-lemma-semi-diagonal (canonical tag 03KO). cited at proof line 3382. Graph closed immersion.

Source reference. Original lemma 03KL: sources/stacks-chow/tags.txt, source lines 4681–4681. Cited at proof line 3377: separatedness stable.

Source reference. Original lemma 03MO: sources/stacks-chow/tags.txt, source lines 4756–4756. Cited at proof line 3378: flatness stable.

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Cited at proof line 3379: properness stable.

Source reference. Original lemma 03XR: sources/stacks-chow/tags.txt, source lines 5158–5158. Cited at proof line 3380: fp stable.

Source reference. Original lemma 03KO: sources/stacks-chow/tags.txt, source lines 4684–4684. Cited at proof line 3382: Gamma_f closed immersion.

Lemma 0.2. *Let S be a scheme. Consider morphisms of algebraic spaces $Z \to B$ and $X \to B$ over S. If $X \to B$ is separated and $Z \to B$ is of finite presentation, flat, and proper, then there is a natural injective transformation of functors*

$$Mor_B(Z, X) \longrightarrow \mathrm{Hilb}_{Z \times_B X/B}$$

which maps a morphism $f : Z_T \to X_T$ to its graph.

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Proof. Given a scheme T over B and a morphism $f_T : Z_T \to X_T$ over T, the graph of f is the morphism $\Gamma_f = (\mathrm{id}, f) : Z_T \to Z_T \times_T X_T = (Z \times_B X)_T$. Recall that being separated, flat, proper, or finite presentation are properties of morphisms of algebraic spaces which are stable under base change (Morphisms of Spaces, Lemmas 03KL, 03MO, 04WP, and 03XR). Hence Γ_f is a closed immersion by Morphisms of Spaces, Lemma 03KO. Moreover, $\Gamma_f(Z_T)$ is flat, proper, and of finite presentation over T. Thus $\Gamma_f(Z_T)$ defines an element of $\mathrm{Hilb}_{Z \times_B X/B}(T)$. To show the transformation is injective it suffices to show that two morphisms with the same graph are the same. This is true because if $Y \subset (Z \times_B X)_T$ is the graph of a morphism f, then we can recover f by using the inverse of $\mathrm{pr}_1|_Y : Y \to Z_T$ composed with $\mathrm{pr}_2|_Y$. $\square$